

```
!pip install chromadb sentence-transformers langchain
```

Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy->onnxruntime==1.14.1-
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch>=1.11.0->sentence-
Requirement already satisfied: oauthlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from requests-oauthlib->kubernetes>
Downloading chromadb-1.5.6-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (22.4 MB)
22.4/22.4 MB 45.0 MB/s eta 0:00:00
Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl (278 kB)
278.2/278.2 kB 15.5 MB/s eta 0:00:00
Downloading build-1.4.2-py3-none-any.whl (24 kB)
Downloading kubernetes-35.0.0-py2.py3-none-any.whl (2.0 MB)
2.0/2.0 MB 48.0 MB/s eta 0:00:00
Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (17.2 MB)
17.2/17.2 MB 59.9 MB/s eta 0:00:00
Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl (20 kB)
Downloading opentelemetry_exporter_otlp_proto_common-1.40.0-py3-none-any.whl (18 kB)
Downloading opentelemetry_proto-1.40.0-py3-none-any.whl (72 kB)
72.1/72.1 kB 2.8 MB/s eta 0:00:00
Downloading opentelemetry_sdk-1.40.0-py3-none-any.whl (141 kB)
142.0/142.0 kB 8.0 MB/s eta 0:00:00
Downloading opentelemetry_api-1.40.0-py3-none-any.whl (68 kB)
68.7/68.7 kB 4.8 MB/s eta 0:00:00
Downloading opentelemetry_semantic_conventions-0.61b0-py3-none-any.whl (231 kB)
231.6/231.6 kB 14.6 MB/s eta 0:00:00
Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (:
71.6/71.6 kB 5.3 MB/s eta 0:00:00
Downloading pypika-0.51.1-py2.py3-none-any.whl (60 kB)
60.6/60.6 kB 4.4 MB/s eta 0:00:00
Downloading durationpy-0.10-py3-none-any.whl (3.9 kB)
Downloading pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Installing collected packages: pypika, durationpy, pyproject_hooks, pybase64, opentelemetry-proto, bcrypt, opentelemetry-exporter
Attempting uninstall: opentelemetry-proto
Found existing installation: opentelemetry-proto 1.38.0
Uninstalling opentelemetry-proto-1.38.0:
Successfully uninstalled opentelemetry-proto-1.38.0
Attempting uninstall: opentelemetry-exporter-otlp-proto-common
Found existing installation: opentelemetry-exporter-otlp-proto-common 1.38.0
Uninstalling opentelemetry-exporter-otlp-proto-common-1.38.0:
Successfully uninstalled opentelemetry-exporter-otlp-proto-common-1.38.0
Attempting uninstall: opentelemetry-api
Found existing installation: opentelemetry-api 1.38.0
Uninstalling opentelemetry-api-1.38.0:
Successfully uninstalled opentelemetry-api-1.38.0
Attempting uninstall: opentelemetry-semantic-conventions
Found existing installation: opentelemetry-semantic-conventions 0.59b0
Uninstalling opentelemetry-semantic-conventions-0.59b0:
Successfully uninstalled opentelemetry-semantic-conventions-0.59b0
Attempting uninstall: opentelemetry-sdk
Found existing installation: opentelemetry-sdk 1.38.0
Uninstalling opentelemetry-sdk-1.38.0:
Successfully uninstalled opentelemetry-sdk-1.38.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is tl
google-adk 1.28.0 requires opentelemetry-api<1.39.0, >=1.36.0, but you have opentelemetry-api 1.40.0 which is incompatible.
google-adk 1.28.0 requires opentelemetry-sdk<1.39.0, >=1.36.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have openteler
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto 1.40.0 wh
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.40.0 wh
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0, >=1.35.0, but you have opentelemetry-sdk 1.40.0
Successfully installed bcrypt-5.0.0 build-1.4.2 chromadb-1.5.6 durationpy-0.10 kubernetes-35.0.0 onnxruntime-1.24.4 openteleme

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
client = chromadb.Client()
collection = client.create_collection(name='rag_collection')
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
modules.json: 100% 349/349 [00:00<00:00, 25.9kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 10.8kB/s]

README.md: 10.5k/? [00:00<00:00, 465kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.95kB/s]

config.json: 100% 612/612 [00:00<00:00, 61.0kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 454MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 367.32it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+---
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 [00:00<00:00, 14.7kB/s]

vocab.txt: 232k/? [00:00<00:00, 9.68MB/s]

tokenizer.json: 466k/? [00:00<00:00, 20.9MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 8.76kB/s]

config.json: 100% 190/190 [00:00<00:00, 15.0kB/s]

```

```

documents = [
    'Artificial Intelligence is the simulation of human intelligence.',
    'Machine Learning is a subset of AI.',
    'Deep Learning uses neural networks.',
    'Natural Language Processing deals with text data.',
    'RAG combines retrieval and generation.'
]
ids = ['doc1', 'doc2', 'doc3', 'doc4', 'doc5']

```

```
embeddings = model.encode(documents)
```

```
collection.add(documents=documents, embeddings=embeddings, ids=ids)
```

```

query = 'What is AI?'
query_embedding = model.encode([query])
results = collection.query(query_embeddings=query_embedding, n_results=2)
print(results)

```

```
{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of human intelligence.
```

```

for doc in results['documents'][0]:
    print(doc)

```

```

Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.

```

```

def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(query_embeddings=query_embedding, n_results=2)
    return results['documents'][0]

print(retrieve('Explain machine learning'))

```

```
['Machine Learning is a subset of AI.', 'Deep Learning uses neural networks.']
```

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with text generation to produce accurate and context-aware responses.